Day 31 – Edge Case & Failure Handling

1. Objective

Day 31 focuses on improving the stability of the AI screening system under real-world edge cases and failure conditions.

The existing screening and conversation architecture was retained. Improvements were focused on speech-to-text processing, conversation fallback behaviour, retry handling, clarification logic, and edge-case validation.

2. Edge Cases Covered

The Day 31 implementation covers the following conditions:

- Poor audio
- Language issues
- Missing responses
- Silence
- Background noise
- Interrupted speech
- Partial responses
- STT provider errors
- Low-confidence speech recognition
- Unknown or unclear responses

3. Existing Edge-Case Baseline

Before making Day 31 changes, the existing related test suite was executed.

Command:

python -m pytest tests/test_speech_to_text_processor.py tests/test_transcript_data_architecture.py tests/test_answer_intent_engine.py tests/test_conversation_flow_engine.py -q

Result:

52 passed in 0.13 seconds

This established a successful baseline for the existing speech, transcript, intent, and conversation-flow components.

4. Speech-to-Text Edge-Case Handling

The existing speech-to-text processor was extended to handle additional real-world conditions.

File updated:

utils/speech_to_text_processor.py

The processor now supports explicit handling for:

- Complete responses
- Interrupted responses
- Partial responses
- Silence
- Poor audio
- Language issues
- Background noise

The processor also identifies interruption and silence markers and provides appropriate retry or clarification requirements.

5. Text Normalization

The speech-to-text processor continues to normalize transcript text by:

- Removing common filler words
- Removing interruption markers
- Normalizing whitespace
- Handling capitalization
- Preserving important technical terms
- Normalizing punctuation

Technical terms such as Python, SQL, API, AI, ML, AWS, Azure, Docker, and Power BI are preserved appropriately during normalization.

6. STT Confidence Handling

A low-confidence threshold was introduced for the speech-to-text integration.

Configured threshold:

0.60

When a completed transcription has confidence below this threshold, it is treated as a poor-audio condition and the system requests a retry.

7. STT Failure Fallback

Safe fallback handling was added for speech-to-text provider failures.

When an STT provider raises an exception, the system returns a controlled fallback response containing:

- Empty transcript
- Zero confidence
- STT error status
- Edge-case indicator
- Retry requirement
- Fallback indicator
- Error information

This prevents an STT provider failure from directly breaking the screening conversation.

8. Conversation Flow Configuration

The existing conversation flow configuration was extended to support Day 31 failure-handling scenarios.

File:

data/conversation_flow_configuration.json

Additional response rules were added for:

- Interrupted responses
- Poor audio
- Language issues
- Background noise
- STT errors

The existing rules for silence, missing or vague responses, off-topic responses, unknown responses, and understood responses were retained.

9. Retry and Clarification Policy

The conversation configuration defines a maximum retry count of:

2 retries

Retry actions include:

- Repeat question
- Clarify question

After the maximum retry count is reached, the configured behaviour is:

continue_with_available_information

This provides a controlled fallback instead of repeatedly asking the candidate the same question.

10. New Edge-Case Tests

Additional Day 31 tests were created to validate the new failure-handling behaviour.

Result:

7 passed in 0.04 seconds

The new tests validated the added edge-case processing and fallback behaviour.

11. STT Regression Testing

After modifying the speech-to-text processor, the existing related tests were executed again.

Result:

52 passed in 0.15 seconds

This confirmed that the Day 31 changes did not break the existing speech, transcript, intent, or conversation-flow functionality.

12. Conversation Flow Regression

The conversation-flow regression tests were executed after updating the configuration.

Result:

52 passed in 0.13 seconds

The existing conversation behaviour remained functional after introducing the new edge-case response rules.

13. Full System Regression

The complete project test suite was executed after the Day 31 changes.

Command:

python -m pytest tests -q

Result:

121 passed, 2 warnings in 15.84 seconds

The full system regression completed successfully.

14. Warnings Observed

The same two existing warnings were reported during the full system test.

14.1 Starlette/httpx Deprecation Warning

A deprecation warning was reported regarding the use of httpx with the Starlette test client.

This warning did not cause test failure.

14.2 Pytest Return Warning

The ATS system evaluation test returns a dictionary instead of using assertions exclusively.

Pytest reported:

PytestReturnNotNoneWarning

This warning did not affect the successful test result.

15. Day 31 Validation Summary

Validation	Result
Existing edge-case baseline	52 passed
New Day 31 edge-case tests	7 passed

Validation	Result
STT regression	52 passed
Conversation flow regression	52 passed
Full system regression	121 passed
Full system warnings	2
STT fallback handling	Passed
Retry handling	Passed
Clarification handling	Passed

16. Existing Architecture

Day 31 does not replace the existing project architecture.

The implementation works with the existing components, including:

- Speech-to-text processing
- Transcript normalization
- Answer intent detection
- Conversation flow
- Screening logic
- Scoring
- Candidate evaluation
- Existing test framework

The changes are limited to improving edge-case and failure handling within the existing system.

17. Day 31 Deliverables

The Day 31 work establishes:

- Robust STT edge-case handling
- Controlled retry behaviour
- Clarification handling
- STT failure fallback
- Poor-audio handling
- Language-issue handling
- Background-noise handling
- Interrupted-response handling
- Edge-case test coverage
- Conversation-flow failure rules
- Full-system regression validation

18. Final Result

Day 31 edge-case and failure-handling implementation was completed successfully.

The updated system successfully passed the new edge-case tests, related regression tests, and the complete existing test suite.

Final full-system result:

121 passed, 2 warnings

No architectural replacement was performed during Day 31.